Assignment 2 (100 marks)

You are given a 3D domain, owned by px X py X pz processes. Each sub-domain size is nx X ny X nz. Iterate over T steps, Dimension of Data[F] per process = nx * ny * nz doubles, F is the number of fields/variables.

Communication: halo exchange for d-point stencil
Computation: d-point stencil (d is input, d < nx, d < ny, d < nz, example values of d are 7, 13, …)
For e.g., average of E,W,N,S,F,B and itself for d=7, i.e. value (P, t+1) = [value (P_W, t) + value (P_E, t) + value (P_N, t) + value (P_S, t) + value (P_F, t) + value (P_B, t) + value (P, t)] / 7
Each process computes the count of isovalues (i.e. the count of points in the domain that would be equal to a given isovalue, refer to Lectures 17 and 18)

Root process calculates the total count of isovalues for every field across all time steps, i.e. there will be T values for field #1, T values for field #2, …. , and T values for field #F.

The initial data must be randomly generated using the value “seed” as shown below.

```
srand(seed);
for (int i=0; i<F; i++)
  for (int j=0; j<arrSize; j++)                  // arrSize = nx * ny * nz
    data[i][j] = (double)rand()*(myrank+1)/(110426.0+i+j);
   // you are free to use any data layout but the initialization should be done like this.
```

The input to the program is d, ppn, px, py, pz, nx, ny, nz, T, random seed, F, isovalue

The output (by rank 0) should be **T+1 lines**, the first T lines contain F numbers, followed by the time:
<count_field1> … <count_fieldF> // for time step 1
…
<count_field1> … <count_fieldF> // for time step T
<time> [put a new line at the end]

Final configurations for the report: (there are a total of 8 configurations)

for execution in 1 to 5 // repeat each configuration 5 times
P (number of processes) in 32, 48, 64, 96 (4 configurations)

if P = 32 then px = 4, py = 4, pz = 2, ppn = 32
if P = 48 then px = 6, py = 4, pz = 2, ppn = 48
if P = 64 then px = 4, py = 4, pz = 4, ppn = 32
if P = 96 then px = 6, py = 4, pz = 4, ppn = 48

nx = ny = nz = 120, nx = ny = nz = 240 (2 configurations)
d = 7
ppn = as per above // For PARAM Rudra (note: this will be set as per the test environment)
T = 5
seed = 1000

F = 2
isovalue = 500

You **must only** use the MPI functions discussed in class. You may optimize your code using concepts learnt in class. Submit your optimized code only.

Plot the time (in seconds) for each data size per process count. Use boxplots (from the 5 executions) for every data point in a plot. Submit one plot file only. Time in seconds on the y-axis and processes (P) on the x-axis.

What to include in the report: Your report must include **succinct** code logic only (not code), your code optimizations and results and the observations from the results. You may include results with and without your code optimizations. **The maximum page limit is 4**.

What to include in the submission: **src.c**, job scripts, plot script and the input (timing data file), SLURM job output files in a sub-folder, **GroupXY.pdf** (report), Makefile (if any). **Include all** your files in a folder named **GroupXY** and then compress it, please name it **GroupXY.zip**.

Submission instructions (***only leaders***): Upload GroupXY.zip to hello.iitk.ac.in. Submissions via email will not be accepted. Multiple submission on MooKIT will incur -5 penalty, so ensure that you have included your final correct code and report in your GroupXY.zip file before uploading.

Approximate split of marks:

Coding (**correct, efficient, compact, concise, clear**) + documentation (**in the code**) = 45
Execution (including hidden test cases) = 40
Report: 15

Penalty of -10 will be given for violation in specified instructions/file naming conventions or omission of any of the submission files.

Optional: You may mention in the report (conclusions) regarding the percentage of work done by every member (only in case of non-uniform workload sharing).

Plagiarism will NOT be tolerated, and CSE/IITK policy will be strictly followed. Inter-group discussions regarding anything even remotely related to the Assignment is not allowed. You must only discuss within your own group and ensure that none of your group members are discussing (not even any simple doubt) with group members of any other group. AI tools are absolutely not allowed to do any part of the assignment, penalty = -100.

Due date: 11:59 PM on 12-04-2026 (Hard deadline)

Late submission (-10 penalty): 11:59 PM on 13-04-2026

Assignment 2 doubts: https://forms.gle/EQuyHDQCsqdSp8NP7

FAQ

Q: Do we double count the boundary isovalue counts?
A: Yes, as discussed in class (after Lecture 19).

Q: Are we searching for the same isovalue (500) for different variables? Variables may have different ranges.
A: Yes, in this assignment we are searching for the same isovalue (input argument) for all variables.

Q: In Assignment 1 we were not allowed to use MPI_Isend and MPI_Irecv. Does the same restriction apply to Assignment 2 as well, or can we use non-blocking communication functions for the halo exchange?
A: Please read the document "You **must only** use the MPI functions discussed in class." MPI_Isend and MPI_Irecv have been covered, hence can be used.

Q: Can we assume for d point stencil computation, the data needed for halo is available with immediate neighboring ranks or do we need to consider the scenario where the data spans multiple neighboring ranks on either direction. For example, do we need to take care of the scenario where each process has 2 x 2 x 2 data and we have to compute 19 point stencil.
A: Yes, that is why the extent of d was given as d < nx, d < ny, d < nz

Q: Should we use a tolerance like fabs(value - isovalue) < 1e-6 instead of exact equality when counting isovalues due to floating-point precision?

A: There is absolutely no need of equating to isovalue in this assignment. The isovalue may lie somewhere in the domain, not on the grid points necessarily. This was explained during the lectures 17 and 18, will be again explained in Lecture 22.

Q: For boundary processes without a neighbor, should we reuse the local value, ignore the missing neighbor, or adjust the denominator by reducing it based on the number of missing stencil contributions? For example, if a grid point normally averages over 6 neighbors (left, right, up, down, front, back) but at the boundary one neighbor is missing, should we: (1) reuse the current value in place of the missing neighbor, (2) compute the average using only the available 5 neighbors, or (3) still divide by 6 but ignore the missing contribution?

A: If the number of neighbours = m, please divide by m, i.e. if there are three neighbours, divide by 3 not 6.

Q: Do we need to take ppn as an input to the source code, since there is no utilization of ppn parameter inside the c code.

A: Yes you need to, that is what is mentioned in the assignment document.

Q: In the assignment, the final configuration for the report has isovalue = 500 (an integer). However, as per the specified data initialization code, our data will be in the form of doubles. So there is negligible chance of our data matching the isovalue at any time instant (499.999... != 500). Hence count will always remain 0. So should we compare only the integer part of data ((int) data == isovalue) OR is a particular range acceptable (data - isovalue < some epsilon)? In the latter option, what should be the range (epsilon)?

A: The assignment does not require equating to isovalue. The isovalue may lie somewhere in the domain, not on the grid points necessarily. Please check with your group members who attended Lectures 17 and 18, I will again explain during Lecture 22.

Q: document states that we need to use only d=7 for final report configs, then why are we taking d as input?

A: For hidden test cases

Q: Is it acceptable to choose a different isovalue that lies within the data range so that crossings occur?

A: Your output will be considered incorrect if you choose a different isovalue

Q: What are the required submission documents

A: Already updated

Q: Is it sufficient to present the timing results in tabular form

A: Please use plot for your main results as mentioned in the instructions. Please note that there are no extra marks for presenting the same result as plots and tables.

Q: for isovalue calculation, as the domain is 3D, standing at grid point we are going to check the East west north south front back points if they meet our iso condition or not , do we need to conside the diagonal points too for this check??

A: See Important FAQ section

Q: While checking for isovalues cell by cell, can a single cell have multiple isovalues? Should a plane passing through a cell be considered a single isovalue? What happens in cases with more than one plane passing through a single cell?

A: We are neither computing isolines, nor isosurfaces.

See Important FAQ section.

Q: Ma'am in the assignment, the rank zero prints the count of isovalues in T iterations. Doubt is that in the first step the isovalue count being printed is this the count taken from the values initialized at the starting or are these the counts after we have done the first stencil computation?

A: Repeat

Halo -> stencil -> count

Q: Do we need to write our code in C, or C++ is fine?

A: Both are fine

Q: what report should contain

A: See the document

Q: for d=13 we select 6 face neighbours one self and 6 edge neighnours out of 12 which 6 do we need to select for this purpose

A: Do not confuse stencil computation with isovalue counting, they are entirely different operations/computations.

Q: Ma'am the doubt is regarding the double counting, when you say we double count the boundary iso-values does this also apply for the intra-process boundaries as in say if within a process two neighbouring grid points have the isovalue between them do we double count it for both these grid points?

A: See Important FAQ section

Q: Maam Is it necessary that we double count the boundary isovalue counts, or is it that we are allowed to double count it but it is not necessary.

A: See Important FAQ section

Q: Ma'am, in today's class you mentioned that we should also consider diagonals for computing isovalues. In this do we also need to consider the body diagonal too apart from face diagonals, i.e do we need to consider all possible diagonals?

A: Diagonal would mean all diagonals (i.e. all neighbours)

See Important FAQ section

Q: In assign 2, to count number of isovalues : we have to use 3D version of marching squares algorithm, i.e. essentially a cubical cell of 2x2x2; or we have to use 3x3x3 cell to count isovalue ?

A: You should not be using marching cubes algorithm; we are not computing isosurfaces. The algorithm to count is pretty simple, just check for values at immediate neighbours, as explained in class.

Q: Do we need to consider diagonal values for isovalue calculations?

A: See Important FAQ section

Q: Should isovalue computation use 26 neighbors or only 6 face neighbors? Also, is double counting limited only to boundary/halo regions between processes, and not present for interior data points?

A: Double counting in case of interior data points is not possible, every information is within a process.

See Important FAQ section

Some important FAQ

Consider all neighbouring grid points (e.g. all diagonal points)?

- I will consider both solutions (i.e. with and without using diagonals), because it seems some of you did not consider diagonal points and have completed the assignment.
- Note, I will not consider a hybrid solution - i.e. be consistent in your code - either consider all the diagonals or do not consider for any case.

Double count the isovalues at process boundaries?

- I will consider both solutions, because it seems some of you are able to avoid that.
- Note, I will not consider a hybrid solution - i.e. your code should not double count for some cases and not for others.

How will the correctness of output be assessed for all test cases?

- Kindly leave it to the instructor. You may be called and asked to explain your code.